RAHUL TRIPATHI

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Summary

Postdoctoral researcher specializing in **computational biology**, **machine learning** and **experimental molecular biology**. Expertise in high-dimensional (**multi-omics**) **data integration**, **spatial transcriptomics**, and **local LLMs** for clinical data extraction. Skilled in characterizing tumor immune microenvironments through WGCNA and spatial gradient analysis, with a solid foundation in **molecular cloning** and **pathway overexpression**.

Skills

- Bioinformatics Analysis
- Omics Integration (Genomics, Transcriptomics, Proteomics, Metabolomics, Epigenetics)
- Single-Cell & Spatial Informatics
- Artificial Intelligence & Machine Learning
- Deep Learning & Large Language Models
- Cheminformatics & Drug Discovery
- Molecular Docking & Dynamics Simulations
- Systems Biology
- Cell Culture
- Recombinant DNA Technology
- Bacterial Transformation
- Molecular Characterization
- Molecular Biology
- Scientific Writing

Experience

Alabama State University, Montgomery, USA — January 2026 to April 2026

Postdoctoral Research Associate

- Deployed **local LLMs (Ollama)** to engineer an automated clinical data extraction pipeline to process 501 unstructured prostate cancer pathology reports from the TCGA and annotate Perineural Invasion (PNI) status to classify
- Executed **differential expression analysis** via **edgeR/limma** and utilized **VST-normalized WGCNA** to identify PNI-driven regulatory modules for downstream targeting.
- Utilized **mixOmics** to **integrate** WGCNA modules, miRNA, epigenetics and protein data to analyze **latent biological drivers** aiming to identify regulatory modules and characterize **molecular mechanisms** underlying PNI in prostate cancer.
- Analyzed **spatial expression gradients** to determine if candidate gene expression correlates with physical proximity to peripheral nerves, exploring potential PNI associations.

Indian Institute of Wheat and Barley Research, Haryana, India — December 2017 to December 2018

Senior Research Fellow

- Contributed to **pipeline development** for identification and genome-wide SSR mining for Loose Smut (*Ustilago segetum tritici*), Covered Smut (*Ustilago hordei*) and Karnal Bunt (*Tilletia indica*).
- Participated in **molecular characterization** of **112 Loose Smut** and **59 Karnal Bunt** isolates through **DNA extraction**, **PCR assays**, and **microscopy** to understand pathogen diversity.
- Conducted **population genetic structure analysis** using STRUCTURE, AMOVA, and **linkage disequilibrium**, contributing to insights on molecular variation and evolutionary dynamics.
- **Mentored 5 trainees** by overseeing technical protocols and providing feedback on workflows, fostering skill development and experimental rigor.

Education

Delhi Technological University, Delhi, India — July 2025

Ph.D. Biotechnology

- **Title of Thesis:** Deciphering Mechanism of Alzheimer's and Parkinson's diseases using network biology and functional genomics approach.
- **Thesis Defense Date:** 26 June 2025
- Identified therapeutic targets by integrating transcriptomic datasets and protein-protein interaction network analysis.
- Evaluated binding stability and pharmacokinetics through virtual screening, 50 ns molecular dynamics simulations (GROMACS), and deep learning-based blood-brain barrier predictions.
- Engineered machine learning models using RDKit-derived molecular descriptors and recursive feature elimination to predict compound inhibitory activity.
- Delivered lectures and supervised lab sessions for undergraduate/graduate students in biotechnology and bioinformatics.
- Mentored 10+ graduate students, overseeing research design and execution in computational biology and molecular modeling.

Motilal Nehru National Institute of Technology, Uttar Pradesh, India — July 2016

M.Tech. Biotechnology (GPA: 8.50)

- Reconstructed a heterologous monolignol biosynthesis pathway by isolating total RNA from Leucaena leucocephala, synthesizing cDNA, and amplifying 4CL, CCR, and CAD genes.
- Constructed recombinant plasmids (pET30b-4CL and pACYCDuet-CCR-CAD) via restriction digestion, T4 DNA ligation, and blue/white screening to enable simultaneous dual-vector expression in E. coli (BL21).
- Executed downstream protein expression protocols, encompassing IPTG induction, cell lysis, and inclusion body extraction for 12% SDS-PAGE analysis.

N.C. College of Engineering (Kurukshetra University), Haryana, India — July 2010

B.Tech. Biotechnology (Score: 71.97%)

- Project 1: Effect of Pesticides on the Growth of Mosquito Larva – A Preliminary Study
- Project 2: Utilization of Whey to Produce Value Added Beverages

Profiles

- LinkedIn: www.linkedin.com/in/rtripathi7
- Google Scholar: www.tinyurl.com/gscholar-rt
- ORCID ID: 0000-0001-5587-0888
- GitHub: www.github.com/rtripathi-compbio

Publications

First-Author Research Publications:

- **Tripathi R**, Kumar P (2025). Identification of CXCR4 Inhibitory Activity in Natural Compounds using Cheminformatics-guided Machine Learning Algorithms. Integrative Biology 17, zyaf004. DOI: 10.1093/intbio/zyaf004
- **Tripathi R**, Kumar P (2023). Preliminary study to identify CXCR4 inhibitors as potential therapeutic agents for Alzheimer's and Parkinson's diseases. Integrative Biology 15, zyad012. DOI: 10.1093/intbio/zyad012

First-Author Review Publication:

- **Tripathi R**, Gupta R, Sahu M, Srivastava D, Das A, Ambasta RK, Kumar P (2022). Free radical biology in neurological manifestations: mechanisms to therapeutics interventions. Environmental Science and Pollution Research 29, 62160–62207. DOI: 10.1007/s11356-021-16693-2

Co-Author Publications:

- Gupta R, Kumari S, **Tripathi R**, Ambasta RK, Kumar P (2023). Unwinding the modalities of necrosome activation and necroptosis machinery in neurological diseases. Ageing Research Reviews 86, 101855. DOI: 10.1016/j.arr.2023.101855
- Gupta R., Sahu M, **Tripathi R**, Ambasta RK, Kumar P (2022). Protein S-sulfhydration: Unraveling the prospective of hydrogen sulfide in the brain, vasculature and neurological manifestations. Ageing Research Reviews 76, 101579. DOI: 10.1016/j.arr.2022.101579
- Sahu M, **Tripathi R**, Jha NK, Jha SK, Ambasta RK, Kumar P (2022). Cross talk mechanism of disturbed sleep patterns in neurological and psychological disorders, Neuroscience & Biobehavioral Reviews 140, 104767. DOI: 10.1016/j.neubiorev.2022.104767
- Advani D, Gupta R, **Tripathi R**, Sharma S, Ambasta RK, Kumar P (2020). Protective role of anticancer drugs in neurodegenerative disorders: A drug repurposing approach. Neurochemistry International 140, 104841. DOI: 10.1016/j.neuint.2020.104841

Co-Author Book Chapter

- Advani D, Sharma S, **Tripathi R**, Gupta R, Jaiswal A, Ambasta RK, Kumar P (2021). Mitochondrial Dysfunction in Metabolic Disorders. Mitochondrial Dysfunction and Nanotherapeutics, 91-137. DOI: 10.1016/B978-0-323-85666-9.00015-2

References

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